Homework 1 · Part B

Training problem set · Lectures 1 and 2

This is the regular assessed homework part, organised as a step-by-step training set. Work at your own pace, use the lecture notes and practise the calculations. Write the submitted solution by hand on paper. There is no time limit for this part.

The assessed total is 100 points, independent of the diagnostic Part A score. The three assessed homeworks are equally weighted within the 20% homework component of the course grade. Cite external sources and disclose any AI assistance; you must be able to explain all submitted work. Submission dates and procedures are announced by the TA.

Submit only B1–B3 below. The additional practice exercises at the end of this assignment are not required, are not graded and do not affect these 100 points.

Show the intermediate steps requested below. In every optimisation problem, distinguish existence, uniqueness and the calculation of a candidate. Unless stated otherwise, quantities are divisible and nonnegative, prices are strictly positive, and income is positive.

B1. A preference checklist · 30 points

On $X = \mathbb{R}^2_+$, consider

$$u_1 = 2x_1 + x_2, \quad u_2 = \min\{x_1, 2x_2\}, \quad u_3 = -(x_1 - 1)^2 - (x_2 - 2)^2.$$

(a) Make a table stating whether each represented relation is complete, transitive and continuous. Explain the common argument behind your answers.

(b) Add columns for convexity of preferences, strict convexity of preferences, strong monotonicity and local nonsatiation. Give one short argument or counterexample for each classification.

(c) For u_2 only, draw the indifference set through $(2, 2)$. Label its utility level and shade its weak upper contour set.

B2. Cobb–Douglas, one step at a time · 35 points

Let $u(x) = x_1^{1/3} x_2^{2/3}$ on $\mathbb{R}^2_+$, with $p_1, p_2, m > 0$.

(a) Establish existence of an optimum using the original utility. Show that every optimum has $x_1, x_2 > 0$ before taking logarithms.

(b) On the positive domain, replace the objective by $g = \log u$. Write its Lagrangian and the first-order conditions for an interior solution with an exhausted budget.

(c) Derive demand for both goods. Verify the budget and both expenditure shares. Explain why this is the unique optimum.

(d) Calculate the bundle at $p = (2, 1)$, $m = 12$. Multiply both prices and income by 3 and check the result.

(e) Derive indirect utility $V_g(p, m)$ for the log objective and use Roy's identity to recover both demands.

B3. Practising corners with Kuhn–Tucker · 35 points

Let $u(x) = 2\log(1 + x_1) + \log(1 + x_2)$, $p = (1, 1)$, $m > 0$.

(a) Write the full KKT system with nonnegative multipliers for $m - x_1 - x_2$, x_1 and x_2.

(b) First assume both goods are positive. Solve the stationarity equations and budget equation. For which incomes is this candidate strictly positive?

(c) Next set $x_2 = 0$, exhaust the budget and calculate all multipliers. For which incomes does dual feasibility hold?

(d) Combine the cases into a demand formula valid for every $m > 0$, including the transition income. Verify global optimality and uniqueness.

(e) Fill a table of $(x_1^*, x_2^*, \lambda, \mu_1, \mu_2)$ for $m = 1/2$, $m = 1$ and $m = 4$. Does $x_2^* = 0$ always imply $\mu_2 > 0$?